CS3230 Lecture 5

NUS National University of Singapore
School of Computing

"Min-Max, Order Statistics, and Linear Time OS"

- **Lecture Topics and Readings**
  - **Linear Time Sorting Algorithms** **[CLRS]-C8.2,8.3**
  - **Min, Max, and Min-Max**
  - **Randomized Divide-and-Conquer** **[CLRS]-C9**
  - **Order Statistics in Linear Time** **[CLRS]-C9**

THINK!

Busting the Lower Bound
Recursive algorithms are elegant!
Balancing leads to efficient algorithms

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MOE Think out of the Box

Singapore
MOE Building
Buona Vista
THINK out of the Box!

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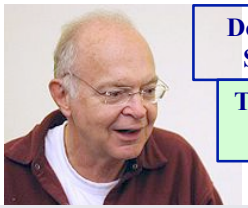
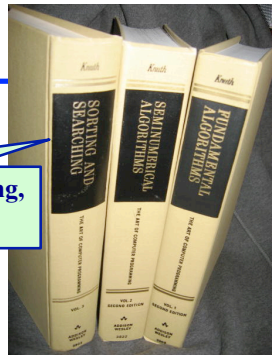
Sorting and Searching



Don Knuth, Stanford

The Art of Computing Programming, Vol 3, "*Sorting and Searching*"



Shen Chuan Qi (2011 SG IOI Team, now at Stanford) attending "Christmas Tree Lecture" given by Don Knuth, around Xmas 2013.

Topic: Planar Graphs and Ternary Trees

Search: Don Knuth, Christmas Tree Lectures, December 2013

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Thank you.

Q & A

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CS3230 Lecture 5(b)

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"Lower Bound for Sorting, Linear-Time Sorting"
"Order Statistics, and Linear Time OS"

❑ **Lecture Topics and Readings**

- ❖ **Order Statistics, Min, Max, Min-Max**
- ❖ **Randomized Divide-and-Conquer [CLRS]-C9**
- ❖ **Order Statistics in Linear Time [CLRS]-C9**

Recursive algorithms are elegant!
Balancing leads to efficient algorithms

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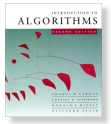

Order statistics

Select the ith smallest of n elements (the element with ***rank*** i).

- $i = 1$: ***minimum***;
- $i = n$: ***maximum***;
- $i = \lfloor (n+1)/2 \rfloor$ or $\lceil (n+1)/2 \rceil$: ***median***.

Naive algorithm: Sort and index ith element.
Worst-case running time $= \Theta(n \lg n) + \Theta(1)$
$= \Theta(n \lg n)$,
using merge sort or heapsort (*not* quicksort).

Quick Revision:
We start with the
humble Find-Max

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Iterative Find-Max algorithm

FIND-MAX $A[1 \,..\, n]$

1. Let $Max\text{-}sf := A[1]$;
2. **for** $k := 2$ **to** n **do**
3. **if** $A[k] > Max\text{-}sf$ **then**
4. $Max\text{-}sf := A[k]$
5. **return** $Max\text{-}sf$

If $A=[3, 1, 5, 7]$ Let $M(x, y) = \text{Max}\ \{x, y\}$

$Max\text{-}sf = 3$, $M(1,3)=3$, $M(5,3)=5$, $M(7,5)=7$

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Iterative Find-Max algorithm

A-Problem:
On average, how often is line 4 executed?

FIND-MAX $A[1 \,..\, n]$

1. Let $Max\text{-}sf := A[1]$;
2. **for** $k := 2$ **to** n **do**
3. **if** $A[k] > Max\text{-}sf$ **then**
4. $Max\text{-}sf := A[k]$
5. **return** $Max\text{-}sf$

Obviously: $T(n) = \Theta(n)$
more precisely: (n–1 comparisons)

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Making Find-Max recursive

FIND-MAX $A[1 \,..\, n]$

1. Let $Max\text{-}sf := A[1]$;
2. **for** $k := 2$ **to** n **do**
3. **if** $A[k] > Max\text{-}sf$ **then**
4. $Max\text{-}sf := A[k]$
5. **return** $Max\text{-}sf$

Compares $A[k]$ with $\max\{A_1, A_2, \ldots, A_{k-1}\}$

Question: ***Can we turn this into a recursive algorithm?***

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Iterative Find-Max algorithm

FIND-MAX $A[1 \,..\, n]$

1. Let $Max\text{-}sf := A[1]$;
2. **for** $k := 2$ **to** n **do**
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5. **return** $Max\text{-}sf$

compares $A[k]$ with $\max\{A_1, A_2, \ldots, A_{k-1}\}$

When k=7, what is value of $Max\text{-}sf$?
$Max\text{-}sf = \max\{A[1], A[2], \ldots A[6]\}$

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Iterative Find-Max algorithm

FIND-MAX $A[1\,..\,n]$

1. Let $Max\text{-}sf := A[1]$;
2. **for** $k := 2$ **to** n **do**
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5. **return** $Max\text{-}sf$

compares $A[k]$ with $\max\{A_1, A_2, \ldots, A_{k-1}\}$

When $k=n$, what is value of $Max\text{-}sf$?…

$Max\text{-}sf = \max\{A[1], A[2], \ldots A[n-1]\}$

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Recursive Find-Max (FMR)

Recursion Schematic:

$\text{FMR}\{A[1..n]\} = \max\{\text{FMR}\{A[1..(n-1)]\},\ A[n]\}$

Find-Max-R $A[1\,..\,n]$

Find-Max-R = FMR

1. If $n = 1$, return $A[1]$
2. $M1$:= Find-Max-R $A[1\,..\,n-1]$
3. return Max $\{A[n], M1\}$

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Recursive Find-Max (FMR)

Find-Max-R $A[1\,..\,n]$

1. If $n = 1$, return $A[1]$
2. $M1$:= Find-Max-R $A[1\,..\,n-1]$
3. return Max $\{A[n], M1\}$

$\text{FMR}\{[3,1,5,7]\}$

$\text{Max}\{7, \text{FMR}\{[3,1,5]\}\}$

If $A=[3, 1, 5, 7]$ }

(Finish this example yourself.
Check with slides at the back.)

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Find-Max and Find-Max-R

Find-Max-R does exactly
the ***same computations*** as Find-Max!

Have the same asymptotic
$\Theta(n)$ worst-case time.

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Recursion Schematics

Recursion Schematic:

$$\boxed{\text{FMR}\{A[1..n]\}} = \max\{\boxed{\text{FMR}\{A[1..(n-1)]\}},\ A[n]\}$$

$(n-1)$ 1

Extreme imbalance

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Recursion Schematics

Recursion Schematic:

$$\boxed{\text{FMR}\{A[1..n]\}} = \max\{\boxed{\text{FMR}\{A[1..(n-1)]\}},\ A[n]\}$$

$(n-1)$ 1

Extreme imbalance

Balanced Recursion Schematic:

$$\text{BFM}\{A[1..n]\}$$
$$= \max\{\ \text{BFM}\{A[1..\ n/2]\}\ ,\ \text{BFM}\{A[n/2+1..\ n]\}\ \}$$

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Balanced Recursive Find-Max

BFM $A[1\ .\ .\ n]$

1. **if** $n = 1$, done.
2. $M1$:= BFM $A[\ 1\ .\ .\ \lceil n/2 \rceil\ ]$
   $M2$:= BFM $A[\ \lceil n/2 \rceil+1\ .\ .\ n\ ]$.
3. **return** max $\{M2, M2\}$

Don't this remind you of Merge-Sort ?

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Recall: Merge sort

MERGE-SORT $A[1 \,..\, n]$

1. If $n = 1$, done.
2. MERGE-SORT $A[\,1 \,..\, \lceil n/2 \rceil\,]$
   MERGE-SORT $A[\,\lceil n/2 \rceil + 1 \,..\, n\,]$
3. "***Merge***" the 2 sorted lists.

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Balanced Recursive Find-Max

BFM $A[1 \,..\, n]$

1. **if** $n = 1$, done.
2. $M1$:= BFM $A[\,1 \,..\, \lceil n/2 \rceil\,]$
   $M2$:= BFM $A[\,\lceil n/2 \rceil + 1 \,..\, n\,]$.
3. **return** max $\{M1, M2\}$

$$T(n) = \begin{cases} \Theta(1) \text{ if } n = 1; \\ 2T(n/2) + \Theta(1) \text{ if } n > 1. \end{cases}$$

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Balanced Recursive Find-Max

BFM $A[1 \,..\, n]$

1. **if** $n = 1$, done.
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   $M2$:= BFM $A[\,\lceil n/2 \rceil + 1 \,..\, n\,]$.
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$$T(n) = \begin{cases} \Theta(1) \text{ if } n = 1; \\ 2T(n/2) + \Theta(1) \text{ if } n > 1. \end{cases}$$

BFM: $a = 2$, $b = 2 \Rightarrow n^{\log_b a} = n^{\log_2 2} = n$
$f(n) = O(n^{1-\varepsilon})$ for $\varepsilon = 0.5 \Rightarrow$ CASE 1: $T(n) = O(n)$.

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Recursion tree

Solve $T(n) = 2T(n/2) + 1$.

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Recursion tree

Solve $T(n) = 2T(n/2) + 1$.

$T(n)$

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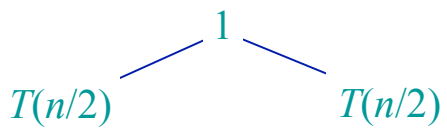
Recursion tree

Solve $T(n) = 2T(n/2) + 1$.



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Recursion tree

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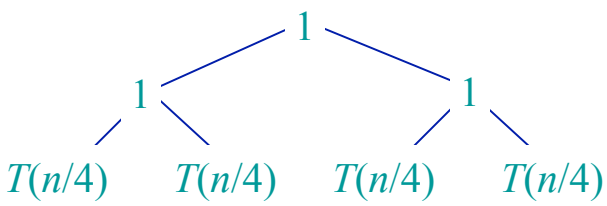

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Recursion tree

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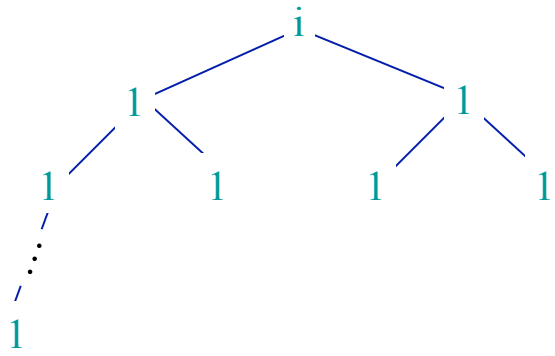

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Recursion tree

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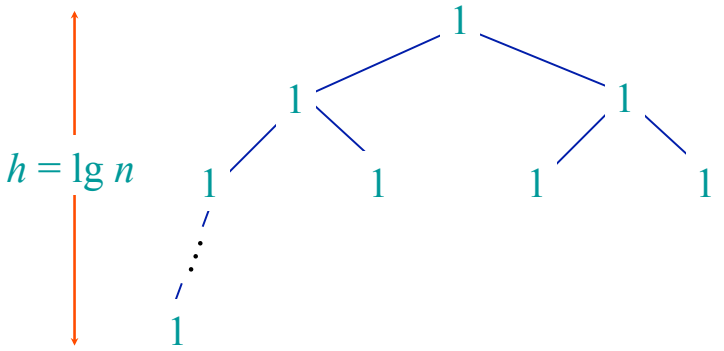

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Recursion tree

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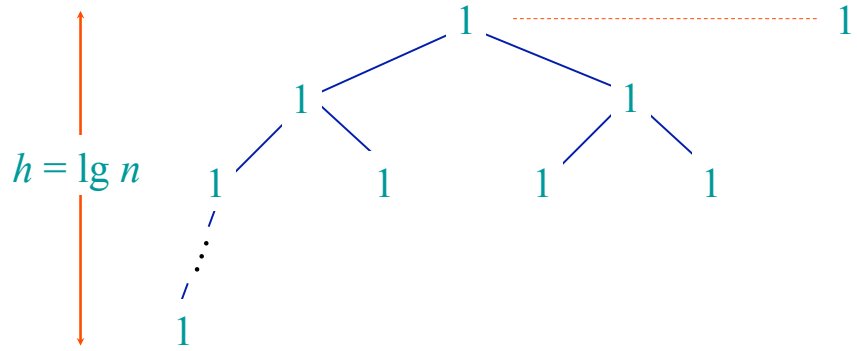

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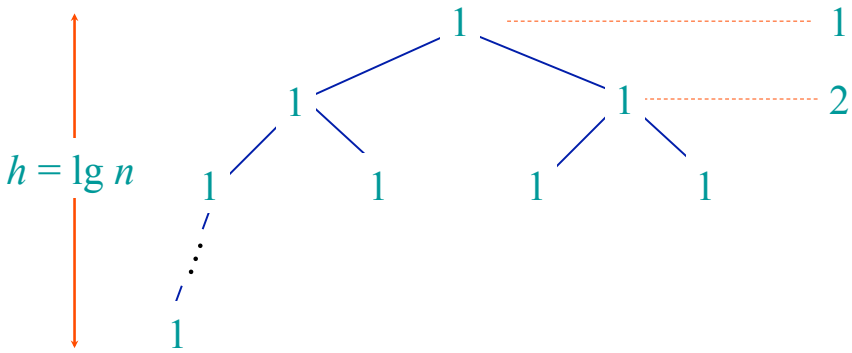

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Recursion tree

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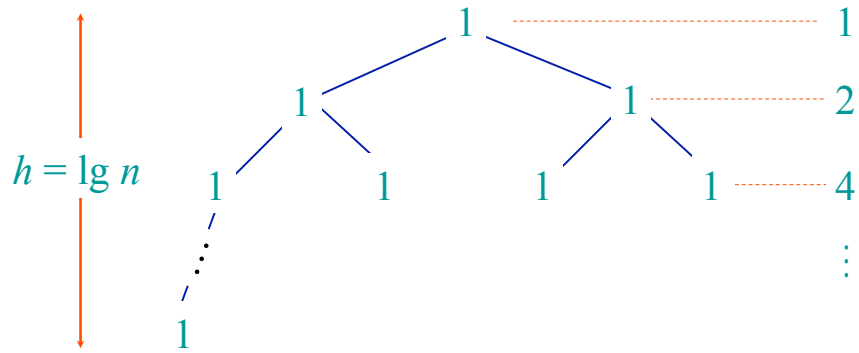

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Recursion tree

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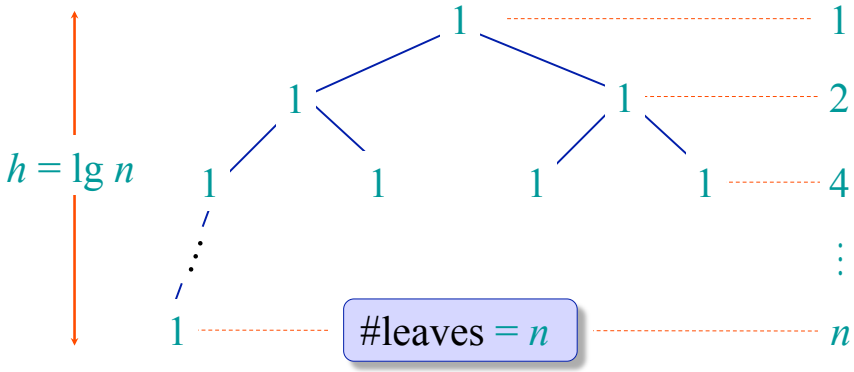

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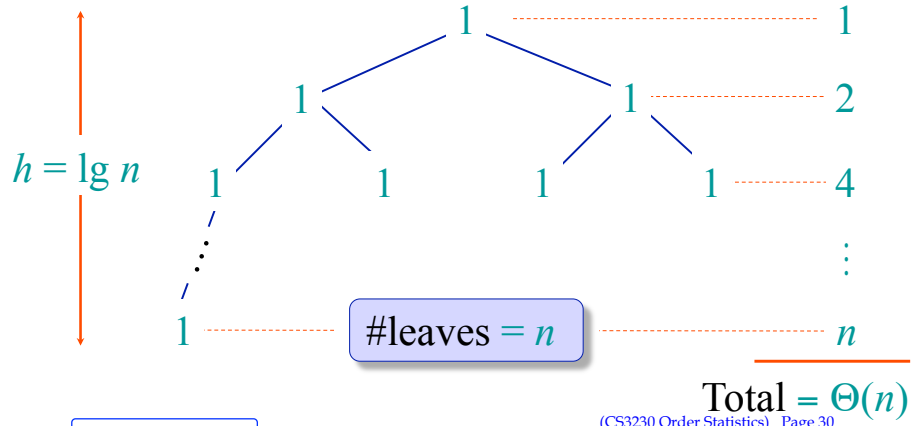
Recursion tree

Solve $T(n) = 2T(n/2) + 1$.



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How to do the sum?

Recall

$$\sum_{k=0}^{\lg n} 2^k = 1 + 2 + 2^2 + \cdots + 2^h \le 2n$$

Or equivalently,

$$\sum_{k=0}^{\lg n} n/2^k = \left(n + n/2 + n/2^2 + \cdots + n/2^{\lg n}\right)$$

$$= n\left(1 + \frac{1}{2} + \frac{1}{2^2} + \ldots + \frac{1}{2^{\lg n}}\right) \le 2n$$

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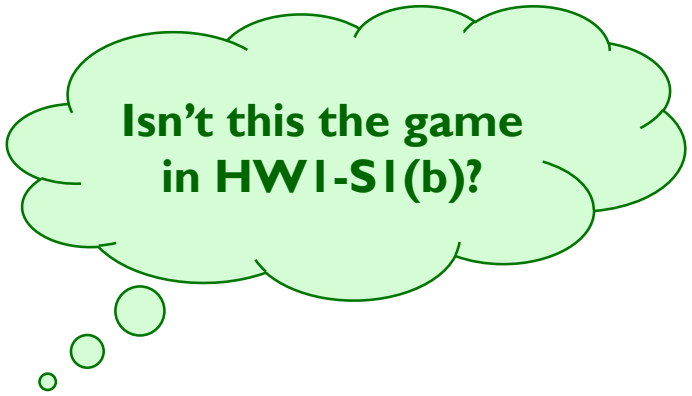
Have you seen this before?



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Modification of HW1-S1(b)

Algorithm (from HW1-S1(b))

1. Each student k stand up, given number A[k]
2. Pair up with someone standing, the one with *smaller number* sits down
3. Go back to Step 2

What's the similarity?

What the difference?

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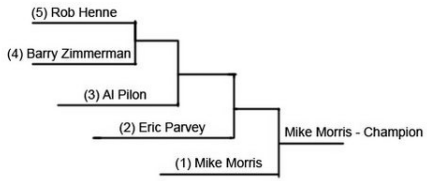
Algorithms is A&E…

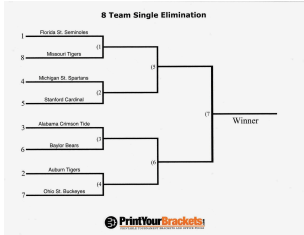
***Recursion Schematic* 1:**

FMR{$A[1..n]$} =
Max{ FMR{$A[1..(n-1)]$} , $A[n]$ }

***Recursion Schematic* 2:**

BFM{$A[1..n]$} =
Max{ BFM{$A[1..\ n/2]$},
BFM{$A[n/2+1..\ n]$} }





Bowling Classic Step-Ladder ***Knock-out Tournament***

http://legacy.wday.com/event/image/id/3122/headline/Bowling%20Classic%20Stepladder/

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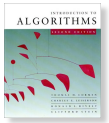

How about finding Max-and-Min ?

Tutorial Question (See T4).

Thank you.

Q & A

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Recursive Find-Max (FMR)

Find-Max-R $A[1 \,..\, n]$

1. If $n = 1$, return $A[1]$
2. $M1$:= Find-Max-R $A[1 \,..\, n-1]$
3. return Max $\{A[n], M1\}$

FMR{[3,1,5,7]}

Max{7, FMR{[3,1,5]} }

Max{5, FMR{[3,1]} }

If A=[3, 1, 5, 7] }

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FMR{[3,1,5,7]}

Max{7, FMR{[3,1,5]} }

Max{5, FMR{[3,1]} }

Max{1, 3 } = 3

If A=[3, 1, 5, 7] }

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Recursive Find-Max (FMR)

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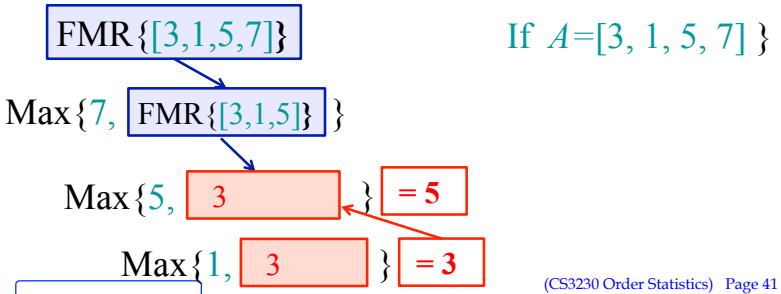

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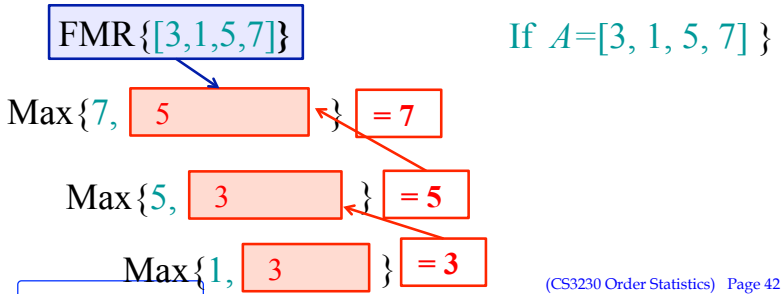

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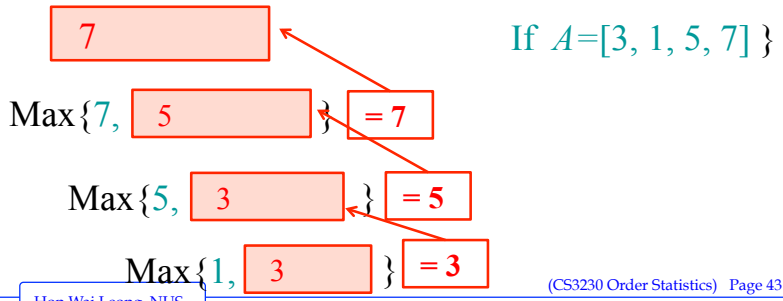

Recursive Find-Max (FMR)

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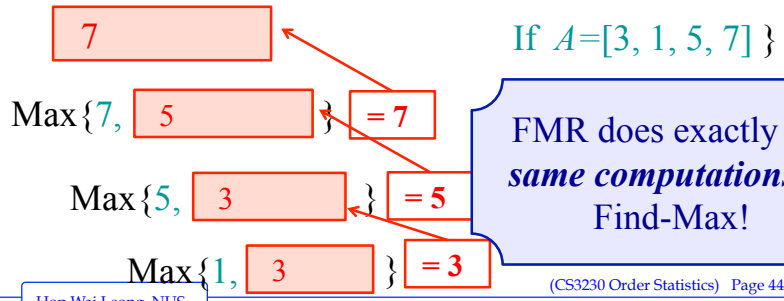

FMR does exactly the ***same computations*** as Find-Max!

CS3230 Lecture 5

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"Linear-Time Sorting"
"Order Statistics, and Linear Time OS"

❑ **Lecture Topics and Readings**

- ❖ Order Statistics, Max, Min-Max [CLRS]-C9.1
- ❖ Randomized Divide-and-Conquer [CLRS]-C9.2
- ❖ Order Statistics in Linear Time [CLRS]-C9.3

Recursive algorithms are elegant!
Balancing leads to efficient algorithms

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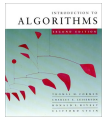

Order statistics

Select the ith smallest of n elements (the element with ***rank*** i).

- $i = 1$: ***minimum***;
- $i = n$: ***maximum***;
- $i = \lfloor (n+1)/2 \rfloor$ or $\lceil (n+1)/2 \rceil$: ***median***.

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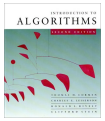
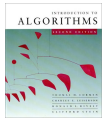
$= \Theta(n \lg n)$,
using merge sort or heapsort (*not* quicksort).



Randomized Divide-and-Conquer Algorithm

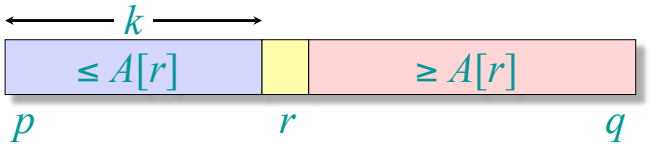
Modified from Quicksort

Also by **C. A. R. (Tony) Hoare**, who invented Quicksort.



Randomized divide-and-conquer algorithm

```
RAND-SELECT(A, p, q, i)        ▷ ith smallest of A[p..q]
  if p = q then return A[p]
  r ← RAND-PARTITION(A, p, q)
  k ← r − p + 1                ▷ k = rank(A[r])
  if i = k then return A[r]
  if i < k
     then return RAND-SELECT(A, p, r − 1, i)
     else return RAND-SELECT(A, r + 1, q, i − k)
```



Example

Select the $i = 7$th smallest:

6	10	13	5	8	3	2	11

$i = 7$

pivot

Partition:

2	5	3	6	8	13	10	11

$k = 4$

Select the $7 - 4 = 3$rd smallest recursively.

Intuition for analysis

(All our analyses today assume that all elements are distinct.)

Lucky:

$T(n) = T(9n/10) + \Theta(n)$ $\quad n^{\log_{(10/9)} 1} = n^0 = 1$

$= \Theta(n)$ $\quad$ CASE 3

Unlucky:

$T(n) = T(n - 1) + \Theta(n)$ $\quad$ arithmetic series

$= \Theta(n^2)$

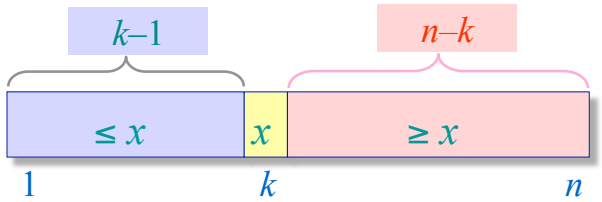
Worse than sorting!

Analysis of RAND-SELECT

Let $T(n)$ = the *expected worst-case* time taken by RAND-SELECT on input of size n.

If pivot x ends up in position k,

then $T(n) = \max\{T(k-1), T(n-k)\} + (n+1)$



Prob(pivot is at pos k) = $1/n$ $\quad$ *for all* k

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Analysis of RAND-SELECT

Then, for expected worst-case, we have

$$T(n) = \begin{cases} \max\{T(0), T(n-1)\} + (n+1) & \text{if } 0 : n-1 \text{ split} \\ \max\{T(1), T(n-2)\} + (n+1) & \text{if } 1 : n-2 \text{ split} \\ \max\{T(2), T(n-3)\} + (n+1) & \text{if } 2 : n-3 \text{ split} \\ \vdots & \vdots \\ \max\{T(n-2), T(1)\} + (n+1) & \text{if } n-2 : 1 \text{ split} \\ \max\{T(n-1), T(0)\} + (n+1) & \text{if } n-1 : 0 \text{ split} \end{cases}$$

Prob(pivot is at pos k) = $1/n$ $\quad$ *for all* k

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Analysis of RAND-SELECT

Then, we have the following recurrence:

$$T(n) = \sum_{k=1}^{n} \frac{1}{n} \cdot \left[\max\{T(k-1), T(n-k)\} + (n+1)\right]$$

Bigger terms appear twice.

$$T(n) \le \frac{2}{n} \sum_{k=\lfloor \frac{n}{2} \rfloor}^{n-1} T(k) + (n+1)$$

$$T(n) \le \frac{2}{n}\left(T(\lfloor \tfrac{n}{2} \rfloor) + T(\lfloor \tfrac{n}{2} \rfloor + 1) + \ldots + T(n-1)\right) + (n+1)$$

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Substitution Method

- We will use "Substitution Method" to prove that $T(n) \le Cn$, for some C.
- **Idea in Substitution Method:**
  1. Guess the form of the solution;
  2. Use mathematical induction to prove it and find the constants
- **Optional for CS3230 (Spring 2014)**
  - **See [CLRS]-C4.3 pp.83-87 for details**

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Using the Substitution Method

$$T(n) \le \frac{2}{n} \sum_{k=\lfloor \frac{n}{2} \rfloor}^{n-1} T(k) + (n+1)$$

Step 1 : Guess $T(n) \le Cn$ for constant $C > 0$.

Step 2: Prove $T(n) \le Cn$ using MI, and find the constant C

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Using the Substitution Method

$$T(n) \le \frac{2}{n} \sum_{k=\lfloor \frac{n}{2} \rfloor}^{n-1} T(k) + (n+1)$$

Prove: $T(n) \le Cn$ for constant $C > 0$.

(Use mathematical induction.)

- **Base Case:** The constant C can be chosen large enough so that $T(n) \le Cn$ for the base cases (n very small).

Later, need fact: $\sum_{k=\lfloor \frac{n}{2} \rfloor}^{n-1} k \le \frac{3}{8} n^2$ (exercise).

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Using the Substitution Method

- **Induction Step:**

$$T(n) \le \frac{2}{n} \sum_{k=\lfloor n/2 \rfloor}^{n-1} Ck + (n+1)$$

Substitute inductive hypothesis.
Namely, $T(k) \le Ck$ for all $k < n$.

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Using the Substitution Method

- **Induction Step:**

$$T(n) \le \frac{2}{n} \sum_{k=\lfloor n/2 \rfloor}^{n-1} Ck + (n+1)$$

$$\le \frac{2C}{n}\left(\frac{3}{8}n^2\right) + (n+1) \qquad \text{(Use fact)}$$

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Using the Substitution Method

- **Induction Step:**

$$T(n) \le \frac{2}{n} \sum_{k=\lfloor n/2 \rfloor}^{n-1} Ck + (n+1)$$

$$\le \frac{2C}{n}\left(\frac{3}{8}n^2\right) + (n+1)$$

$$= Cn - \left(\frac{Cn}{4} - (n+1)\right)$$

Express as ***desired – residual***.

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Using the Substitution Method

- **Induction Step:**

$$T(n) \le \frac{2}{n} \sum_{k=\lfloor n/2 \rfloor}^{n-1} Ck + (n+1)$$

$$\le \frac{2C}{n}\left(\frac{3}{8}n^2\right) + (n+1)$$

$$= Cn - \left(\frac{Cn}{4} - (n+1)\right)$$

When C=5, then
$(Cn/4 - (n+1))$
$= (n/4 - 1) \ge 0$ for $n \ge 4$.

$$\le Cn \qquad \text{(end of induction proof)}$$

Choose C=5, $n_0 = 4$,
then residual term ≥ 0 for $n > n_0$.

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Summary of randomized order-statistic selection

- Works fast: linear expected time.
- Excellent algorithm in practice.
- But, the worst case is ***very*** bad: $\Theta(n^2)$.

Q. Is there an algorithm that runs in linear time in the worst case?

A. Yes, due to Blum, Floyd, Pratt, Rivest, and Tarjan [1973].

IDEA: Generate a good pivot recursively.

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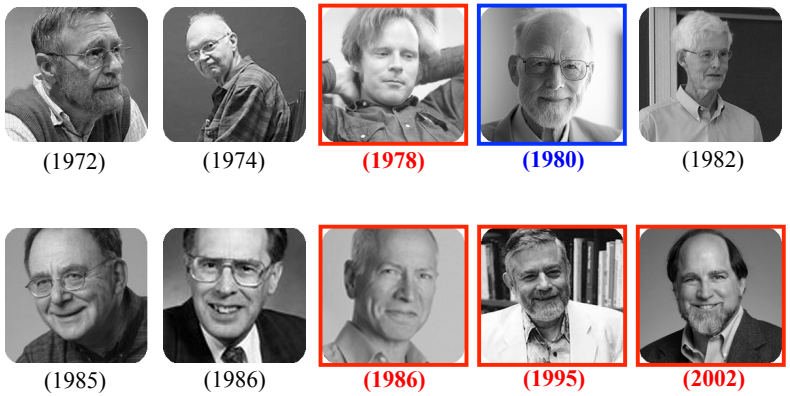
Worst-case Linear-Time Order Statistic Algorithm

M. Blum, R. W. Floyd, V. R. Pratt, R. L. Rivest, R. E. Tarjan, "Time Bounds for Selection," Journal of Computer and System Sciences, (Aug 1973), **7** (4): 448–461. doi:10.1016/S0022-0000(73)80033-9

Why is CS3230 FUN?

- "Meet" many CS celebrities



Worst-case linear-time order statistics

SELECT(i, n)

1. Divide the n elements into groups of 5. Find the median of each 5-element group by rote.
2. Recursively SELECT the median x of the $\lfloor n/5 \rfloor$ group medians to be the pivot.
3. Partition around the pivot x. Let $k = \text{rank}(x)$.
4. **if** $i = k$ **then return** x
   elseif $i < k$
   then recursively SELECT the ith smallest element in the lower part
   else recursively SELECT the $(i-k)$th smallest element in the upper part

Same as RAND-SELECT

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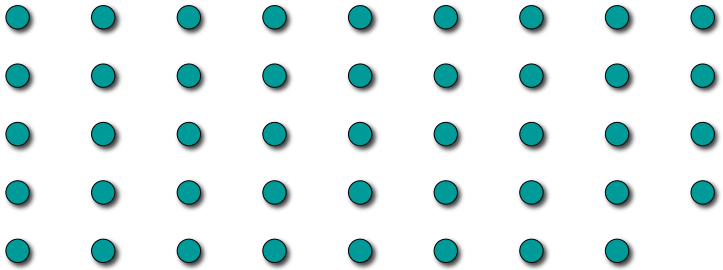
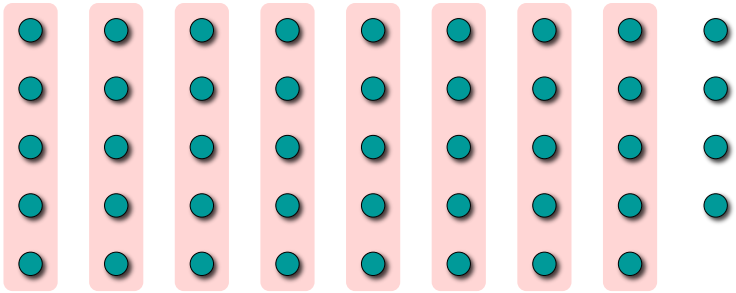
Choosing the pivot

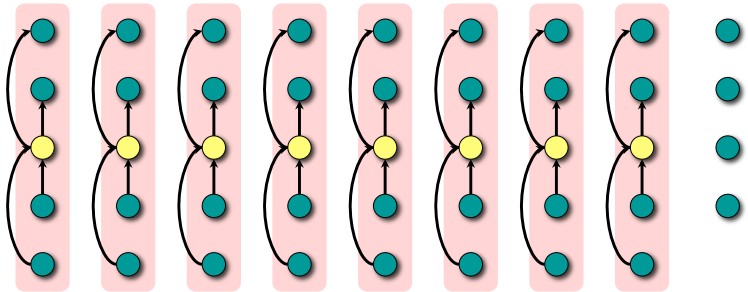


Choosing the pivot

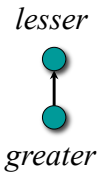


1. Divide the n elements into groups of 5.

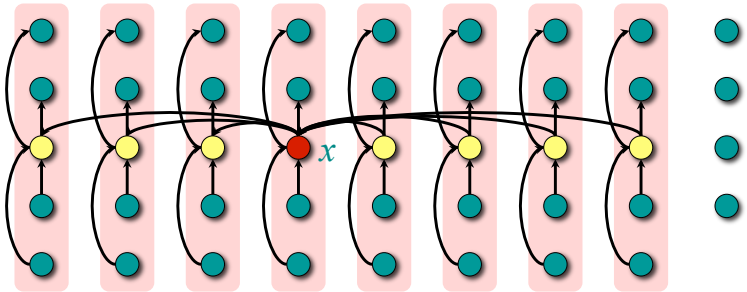
Choosing the pivot



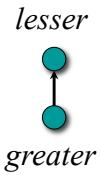
1. Divide the n elements into groups of 5. Find the median of each 5-element group by rote.



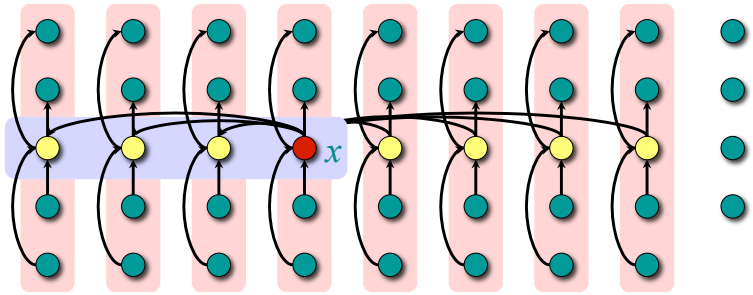
Choosing the pivot



1. Divide the n elements into groups of 5. Find the median of each 5-element group by rote.
2. Recursively SELECT the median x of the $\lfloor n/5 \rfloor$ group medians to be the pivot.



Analysis



At least half the group medians are $\le x$, which is at least $\lfloor \lfloor n/5 \rfloor /2 \rfloor = \lfloor n/10 \rfloor$ group medians.

lesser

greater

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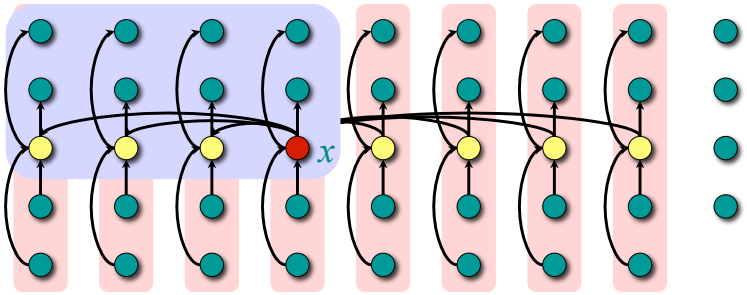
Analysis (Assume all elements are distinct.)



At least half the group medians are $\le x$, which is at least $\lfloor \lfloor n/5 \rfloor /2 \rfloor = \lfloor n/10 \rfloor$ group medians.

- Therefore, at least $3\lfloor n/10 \rfloor$ elements are $\le x$.

lesser

greater

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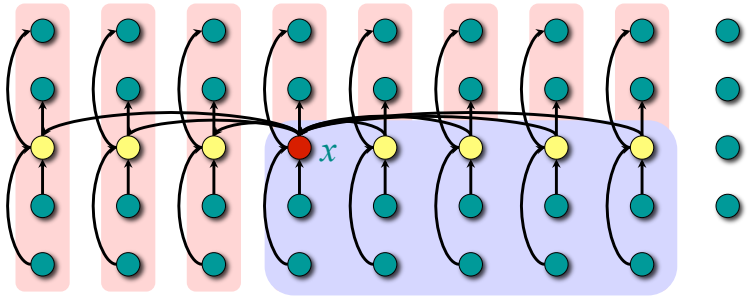
Analysis (Assume all elements are distinct.)



At least half the group medians are $\le x$, which is at least $\lfloor \lfloor n/5 \rfloor /2 \rfloor = \lfloor n/10 \rfloor$ group medians.

- Therefore, at least $3\lfloor n/10 \rfloor$ elements are $\le x$.
- Similarly, at least $3\lfloor n/10 \rfloor$ elements are $\ge x$.

lesser

greater

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Minor simplification

- For $n \ge 50$, we have $3\lfloor n/10 \rfloor \ge n/4$.
- Therefore, for $n \ge 50$ the recursive call to SELECT in Step 4 is executed recursively on $\le 3n/4$ elements.
- Thus, the recurrence for running time can assume that Step 4 takes time $T(3n/4)$ in the worst case.
- For $n < 50$, we know that the worst-case time is $T(n) = \Theta(1)$.

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Developing the recurrence

$T(n)$ SELECT(i, n)

$\Theta(n)$ 1. Divide the n elements into groups of 5. Find the median of each 5-element group by rote.

$T(n/5)$ 2. Recursively SELECT the median x of the $\lfloor n/5 \rfloor$ group medians to be the pivot.

$\Theta(n)$ 3. Partition around the pivot x. Let $k = \text{rank}(x)$.

$T(3n/4)$ 4. **if** $i = k$ **then return** x
elseif $i < k$
then recursively SELECT the ith smallest element in the lower part
else recursively SELECT the $(i-k)$th smallest element in the upper part

Solving the recurrence

$$T(n) = T\left(\frac{1}{5}n\right) + T\left(\frac{3}{4}n\right) + \Theta(n)$$

Substitution: $T(n) \le cn$

$$\begin{aligned} T(n) &\le \frac{1}{5}cn + \frac{3}{4}cn + \Theta(n) \\ &= \frac{19}{20}cn + \Theta(n) \\ &= cn - \left(\frac{1}{20}cn - \Theta(n)\right) \\ &\le cn \quad , \end{aligned}$$

if c is chosen large enough to handle both the $\Theta(n)$ and the initial conditions.

Conclusions

- Since the work at each level of recursion is a constant fraction (19/20) smaller, the work per level is a geometric series dominated by the linear work at the root.
- In practice, this algorithm runs slowly, because the constant in front of n is large.
- The randomized algorithm is far more practical.

Exercise: *Why not divide into groups of 3?*